

SAVUKOSKI KK, Finland

WMO: 028150

Lat: 67.286N

Lon: 28.177E

Elev: 180

StdP: 99.18

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-28.8	-35.1	0.1	-31.9	-31.7	0.2	-28.7	5.5	-6.1	4.8	-3.7	0.0	0	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	25.6	16.5	23.1	15.6	21.0	14.8	18.1	23.0	16.9	21.6	15.8	19.5	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	12.0	19.6	15.1	11.0	18.1	14.0	10.2	17.2	51.7	22.9	48.1	21.7	44.7	19.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	4.1	3.5	DB	-35.5	26.9	3.6	2.2	-38.0	28.5	-40.1	29.7	-42.1	31.0	-44.7	32.6	(4)
(5)			WB	-35.4	19.8	3.4	1.6	-37.8	21.0	-39.8	21.9	-41.7	22.7	-44.2	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-13.4	-11.7	-7.6	-0.8	6.2	10.9	14.7	12.1	7.4	0.0	-5.3	-9.0	(6)
(7)		DBStd	10.80	7.37	8.05	5.91	4.41	4.59	4.23	3.70	3.45	3.27	4.31	5.55	6.90	(7)
(8)		HDD10.0	3844	725	607	544	323	136	38	5	18	90	311	459	588	(8)
(9)		HDD18.3	6571	983	840	803	573	376	225	121	197	329	569	709	847	(9)
(10)		CDD10.0	330	0	0	0	0	19	66	152	82	11	0	0	0	(10)
(11)		CDD18.3	14	0	0	0	0	0	3	9	2	0	0	0	0	(11)
(12)		CDH23.3	189	0	0	0	0	12	39	112	26	0	0	0	0	(12)
(13)		CDH26.7	32	0	0	0	0	1	6	21	5	0	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.0	1.2	1.6	1.6	1.8	1.7	1.5	1.4	1.4	1.3	1.2	1.2	(14)
(15)	Precipitation	PrecAvg	593	39	36	31	33	46	66	83	63	55	55	46	41	(15)
(16)		PrecMax	774	79	73	65	79	85	110	137	121	112	109	71	70	(16)
(17)		PrecMin	444	14	14	7	8	10	17	18	14	9	21	16	11	(17)
(18)		PrecStd	83	15	17	15	16	17	27	31	27	26	24	17	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	2.4	5.4	14.0	25.0	27.6	29.2	26.8	18.1	9.8	5.1	2.3	(19)
(20)			MCWB	0.0	1.6	2.2	7.6	13.7	17.1	18.0	19.0	13.6	8.9	4.6	1.8	(20)
(21)		2%	DB	-1.0	0.8	3.1	8.7	21.4	23.6	26.2	22.7	16.2	8.1	3.2	0.9	(21)
(22)			MCWB	-1.4	0.3	0.7	4.1	12.6	15.6	16.8	16.1	12.2	6.9	2.8	0.6	(22)
(23)		5%	DB	-2.3	0.0	1.5	6.8	17.4	20.7	24.0	20.4	14.4	6.8	1.8	0.2	(23)
(24)			MCWB	-2.6	-0.4	-0.1	3.1	10.5	13.6	16.3	15.2	11.7	5.9	1.6	0.0	(24)
(25)		10%	DB	-3.9	-1.4	0.4	5.2	14.4	18.3	21.9	18.3	12.7	5.5	0.8	-0.7	(25)
(26)			MCWB	-4.2	-1.8	-0.7	2.0	9.0	12.6	15.7	14.2	10.8	4.8	0.6	-0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.7	2.5	7.5	14.5	18.3	20.4	19.8	14.6	9.4	5.1	1.8	(27)
(28)			MCDWB	0.5	2.3	4.6	13.3	22.8	24.6	24.6	25.3	17.1	9.8	5.1	2.2	(28)
(29)		2%	WB	-1.3	0.5	1.2	4.6	13.0	16.7	18.5	17.3	13.0	7.1	2.8	0.7	(29)
(30)			MCDWB	-1.0	0.8	2.6	8.0	20.9	22.6	22.8	21.4	15.1	7.8	3.1	0.8	(30)
(31)		5%	WB	-2.6	-0.4	0.3	3.4	11.3	14.9	17.4	15.9	12.1	5.9	1.6	0.1	(31)
(32)			MCDWB	-2.3	0.0	1.2	6.0	16.8	18.8	22.0	19.1	13.9	6.7	1.8	0.2	(32)
(33)		10%	WB	-4.2	-1.8	-0.5	2.5	9.3	13.4	16.4	14.7	11.0	4.9	0.7	-0.9	(33)
(34)			MCDWB	-3.9	-1.4	0.4	4.6	13.6	17.3	20.7	17.3	12.5	5.5	0.8	-0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.0	10.8	9.3	9.8	9.2	9.7	9.3	7.5	5.0	5.1	6.7	(35)
(36)			MCDBR	7.9	6.2	8.3	10.9	15.8	13.1	14.1	12.9	10.7	5.9	4.2	5.0	(36)
(37)			MCWBR	7.8	5.9	6.4	6.7	8.0	6.1	6.1	6.8	7.0	4.9	4.2	4.9	(37)
(38)		5% WB	MCDBR	8.1	6.0	6.3	9.4	14.5	11.2	11.6	10.9	9.1	5.1	4.2	4.8	(38)
(39)			MCWBR	8.1	5.7	5.1	5.9	7.8	6.0	6.0	6.2	6.4	4.6	4.2	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.193	0.257	0.292	0.323	0.341	0.343	0.356	0.337	0.311	0.278	0.352	N/A	(40)	
(41)		taud	1.923	2.090	2.152	2.195	2.406	2.469	2.451	2.477	2.533	2.322	2.708	N/A	(41)	
(42)		Ebn at Noon	152	558	745	824	856	864	837	816	755	550	184	0	(42)	
(43)		Edh at Noon	12	51	83	106	97	93	92	81	60	41	14	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.06	0.54	2.02	3.80	4.80	5.03	4.61	3.34	1.76	0.57	0.11	0.00	(44)	
(45)		RadStd	0.00	0.08	0.17	0.27	0.46	0.41	0.48	0.32	0.17	0.10	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page